Performance Testing

Date	5 July 2026
Team ID	xxxxxx
Project Name	Emotion Detection and Learning Support Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Manual Testing, Browser DevTools, Postman
Type of Testing	Functional Testing, API Testing, Basic Performance Testing
Target Module / API	Emotion Detection API (/predict), Dashboard, Login, Camera Module
Test Environment	Local Machine / Render Cloud Server
Test Date	5 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Emotion Detection API	1	60 sec	Correct emotion prediction
2	User Login & Dashboard	1	60 sec	Successful login and dashboard loading
3	Camera Capture & Prediction	1	60 sec	Image captured and emotion displayed

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Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	~1.2 sec	Pass	Fast response
2	Response Time (Max)	< 5 seconds	~2.4 sec	Pass	Within limit
3	Throughput (Req/sec)		Basic manual testing	Pass	No issues
4	Error Rate	< 1%	0%	Pass	No errors during testing
5	CPU Utilization	< 80%	~35%	Pass	Normal
6	Memory Utilization	< 80%	~45%	Pass	Stable

Step 4: Observations & Analysis

Key Findings

- Application responds quickly.
- Emotion detection works correctly.
- Dashboard loads without noticeable delay.

Bottlenecks Identified

- Initial backend wake-up delay on Render (if inactive).
- Camera permission required on first use.

Optimization Steps Taken

- Optimized API calls using Axios.
- Modular frontend components.
- Efficient FastAPI backend implementation.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):